3 Lightning occurs when charge builds up in the atmosphere, creating a potential difference between the ground and the atmosphere.

During a lightning strike there is an average current of 3.3×10^4 A for a time of 2.6×10^{-5} s.

(a) Calculate the charge transferred during the lightning strike.

charge = ... C [2]

(b) The potential difference between the ground and the atmosphere is 3.0×10^7 V.

Calculate the average power, in GW, transferred during the lightning strike.

power = ... GW [2]

(c) A lightning rod is attached to a tall building to conduct charge safely to the ground. The lightning rod is modelled as a uniform cylindrical copper cable of total length 95 m that runs from the ground to the top of the building, as shown in Fig. 3.1.

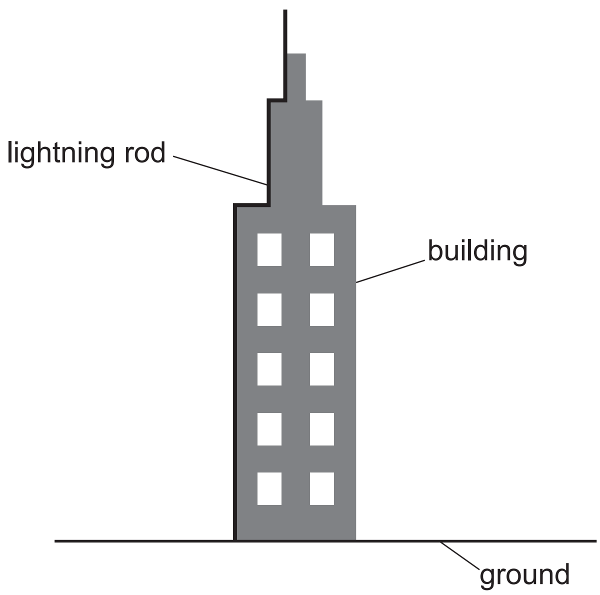

Fig. 3.1

(i) The resistance of the lightning rod is $9.6\,\Omega$.
The resistivity of copper is $1.7 \times 10^{-8}\,\Omega\,\text{m}$.

Determine the radius of the lightning rod.

radius = ... m [3]

(ii) The radius of the copper lightning rod is doubled with no change to its length.

State the effect of this change on the resistance of the lightning rod.

.... [1]

(d) A section of the lightning rod of length 0.12 m is removed for testing. A tensile stress of 1.9×10^{6} Pa is applied, as shown in Fig. 3.2.

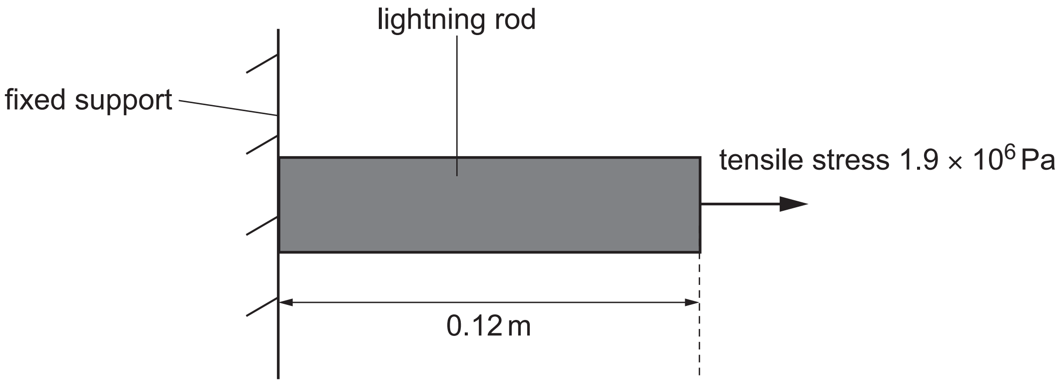

Fig. 3.2 (not to scale)

The section of the rod obeys Hooke's law. The Young modulus of copper is 1.3×10^{11} Pa.

Calculate the extension of the section.

extension = ... m [3]

[Total: 11]